

MATHEMATICS ASSIGNMENT

**A REPORT ON THE ACTIVITIES AND
FUNCTIONS**

OF STATISTICAL DEPARTMENTS

**UNDER THE GOVERNMENT OF INDIA AND
KERALA**

Ministry of Statistics and Programme Implementation
Government of India

and

Department of Economics and Statistics
Government of Kerala

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Course: M.Tech CSE (AI & SE)

1. Introduction

Statistics plays an important role in government administration. It provides information for planning, policy formulation, resource allocation and evaluation of development programmes. Statistical departments collect, analyse and publish economic, social, demographic and environmental data.

In India, the major central institution responsible for official statistics is the **Ministry of Statistics and Programme Implementation (MoSPI)**. At the state level, the **Department of Economics and Statistics (DES), Government of Kerala** performs major statistical functions.

2. Statistical Activities under the Government of India

MoSPI is the nodal ministry for the organized development of India's statistical system. Its activities are mainly carried out through the National Statistical Office and other divisions.

The major activities and functions include:

- Conducting sample surveys on employment, consumption, health, education and socio-economic subjects.
- Preparing National Accounts Statistics, including GDP, GVA and related indicators.
- Compiling price statistics such as the Consumer Price Index (CPI).
- Preparing industrial statistics, including IIP and ASI.
- Producing labour and employment statistics through the Periodic Labour Force Survey (PLFS).
- Developing indicators for monitoring Sustainable Development Goals (SDGs).
- Disseminating statistical reports and datasets.
- Monitoring selected government programmes and infrastructure projects.

These activities provide quantitative evidence for national planning and public policy.

3. Statistical Activities under the Government of Kerala

The **Department of Economics and Statistics (DES), Government of Kerala** is the nodal agency for official statistics in the state. It collects and analyses data required for state planning and development.

The major activities and functions include:

- Preparing State Income Statistics, including GSDP, GVA and per-capita income.

- Collecting agricultural statistics on crop area, production, yield and land utilisation through activities such as EARAS.
- Collecting price statistics, including retail prices and consumer price indices.
- Preparing industrial statistics through IIP and ASI.
- Collecting population and vital statistics, including births and deaths.
- Conducting and participating in National Sample Surveys on employment, household expenditure, health and education.
- Compiling labour and housing statistics.
- Providing local-level statistics for district, municipality and panchayat planning.

4. Recent Statistical Survey: PLFS

The recent **Periodic Labour Force Survey (PLFS) 2023–24** provides a useful example of statistical activity in both India and Kerala. For persons aged 15 years and above, the unemployment rate was **3.2% in India** and **7.2% in Kerala**. The following graph shows the unemployment trend reported through PLFS from 2017–18 to 2023–24.

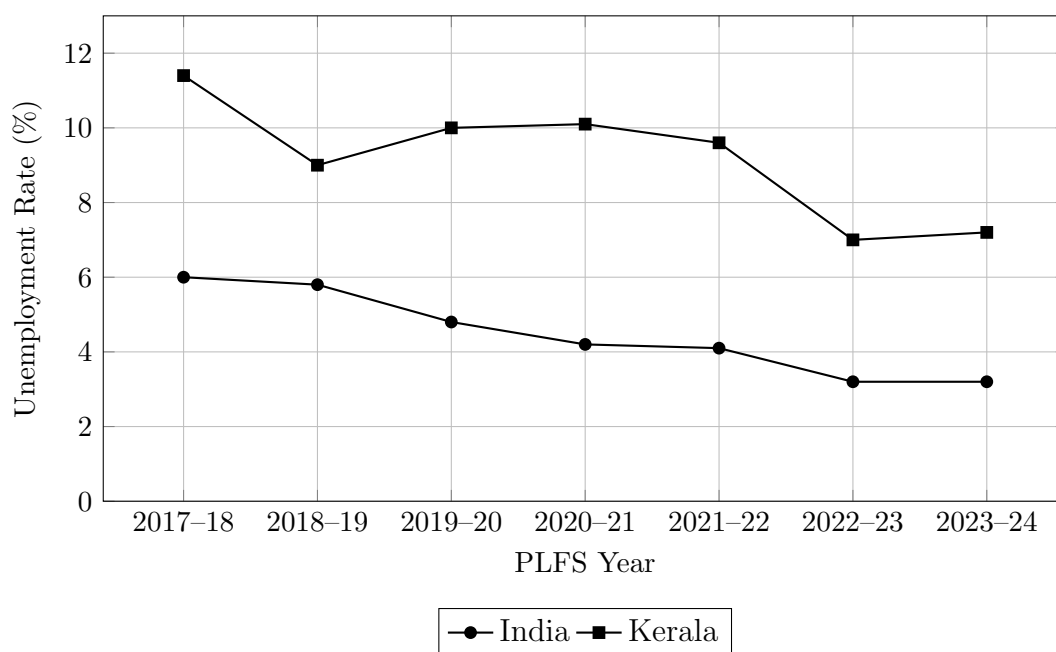


Figure 1: Unemployment Rate in India and Kerala, PLFS 2017–18 to 2023–24

5. Census and Recent Survey Data

The **Population Census 2011** remains the latest completed population census for India and Kerala. Selected indicators show differences in population, literacy and sex ratio between the national and state levels.

Indicator	India	Kerala
Population (million)	1210.9	33.4
Literacy Rate (%)	74.04	93.91
Sex Ratio (females per 1000 males)	943	1084

Table 1: Selected Census 2011 Indicators for India and Kerala

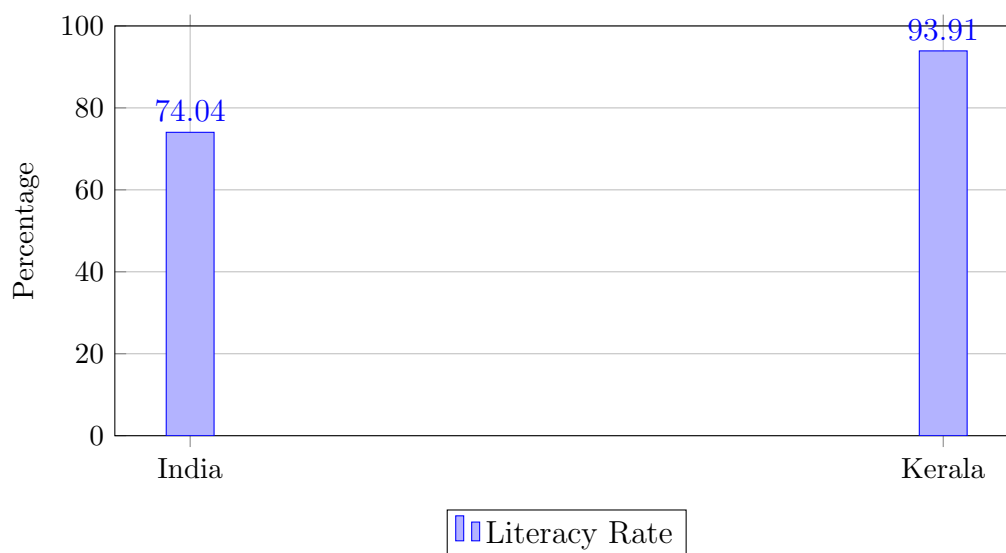


Figure 2: Literacy Rate in India and Kerala, Census 2011

5.1 Household Consumption Expenditure Survey 2022–23

The **Household Consumption Expenditure Survey (HCES) 2022–23** provides recent information on average monthly per-capita consumption expenditure (MPCE). The figures below compare rural and urban India with Kerala.

Area	Rural MPCE (Rs.)	Urban MPCE (Rs.)
All-India	3,773	6,459
Kerala	5,924	7,078

Table 2: Average Monthly Per-Capita Consumption Expenditure, HCES 2022–23

6. Comparison of Major Functions

7. Importance of Government Statistics

Government statistics supports budgeting, employment planning, agricultural development, industrial policy, welfare programmes and infrastructure development.

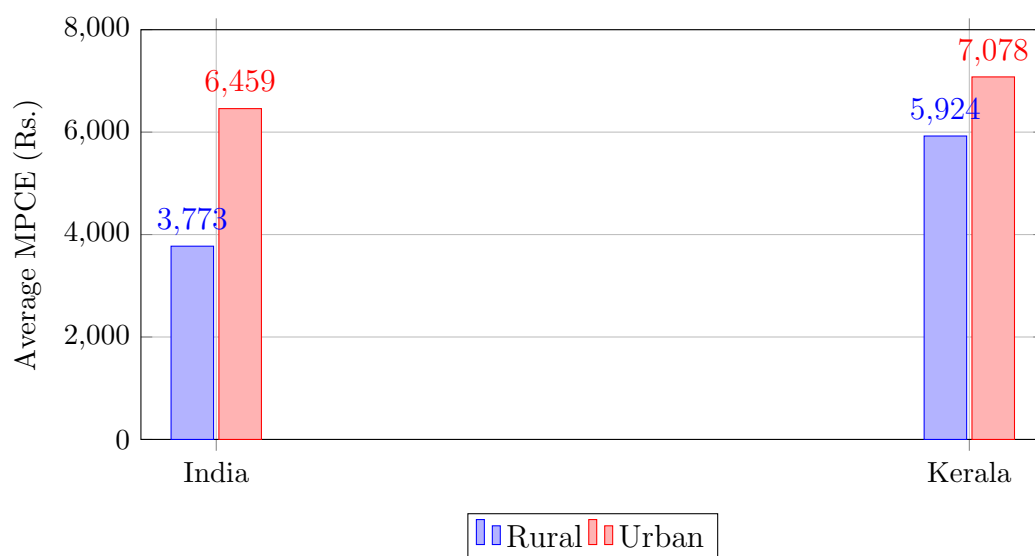


Figure 3: Average MPCE in India and Kerala, HCES 2022–23

Area	Government of India	Government of Kerala
Main Institution	MoSPI / National Statistical Office	Department of Economics and Statistics
Economic Statistics	GDP, GVA and National Accounts	GSDP, GVA and State Income
Agriculture	National agricultural statistics	EARAS and agricultural statistics
Industry	IIP, ASI and Economic Census	IIP, ASI and industrial statistics
Prices	CPI and inflation statistics	CPI, retail and other price statistics
Labour	PLFS and labour surveys	Labour and housing statistics
Socio-economic Surveys	National Sample Surveys	State and National Sample Surveys
Demography	National demographic statistics	State-level vital statistics

Table 3: Comparison of Statistical Activities in India and Kerala

Mathematical methods such as sampling, estimation, averages, variance, index numbers, correlation and regression are widely used. Thus, statistical departments provide an important practical application of mathematics and statistics.

8. Conclusion

The statistical departments of India and Kerala support evidence-based governance. MoSPI produces national statistics, while DES performs similar functions at the state level, with emphasis on agriculture, state income, local statistics and socio-economic conditions.

Reliable statistics enables governments to identify development needs, formulate policies and measure outcomes. Thus, statistical departments connect mathematical analysis with government administration and social development.

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